Anand Bhattad

ADDRESS

3340 Thomas M. Siebel Center
201 N. Goodwin Avenue
Urbana, Illinois 61801, USA

Voice: (217) 904-5911
E-mail: anandbhattad92@gmail.com
web: https://anandbhattad.github.io/

EDUCATION

University of Illinois, Urbana-Champaign (UIUC), USA

Ph.D., Computer Science — *Jan'19 - Present*
Advisor: David Forsyth
Thesis committee: David Forsyth (chair), Alyosha Efros, Bill Freeman, Derek Hoiem, Shenlong Wang and Svetlana Lazebnik

M.S., Computer Science — *Aug'17 - Dec'18*
M.S., Civil Engineering — *Aug'15 - Aug'17*

National Institute of Technology Karnataka (NITK) Surathkal, India

B.Tech., Civil Engineering — *July'11 - May'15*

REFERENCES

David Forsyth, Professor, Computer Science, UIUC – daf@illinois.edu
Derek Hoiem, Professor, Computer Science, UIUC – dhoiem@illinois.edu
Shenlong Wang, Assistant Professor, Computer Science, UIUC – shenlong@illinois.edu
Yuxiong Wang, Assistant Professor, Computer Science, UIUC – yxw@illinois.edu

PREPRINTS /UNDER REVIEW

6. **StyLitGAN: Prompting StyleGAN for Generating Images with New Illumination Conditions**
A. Bhattad, D. A. Forsyth
In Submission

5. **Illuminator: Learning to Correct Lighting Without Ground Truth Data**
A. Bhattad, B.Chen, S. Wang, S. Richter, D. A. Forsyth
In Submission

4. **Make It So: Steering StyleGAN for Any Image Inversion and Editing**
A. Bhattad, V. Shah, D. Hoiem, D. A. Forsyth
In Submission

3. **Learning Equivariant Dense Prediction Networks**
Yuanyi Zhong, **A. Bhattad**, Y. Wang, D. A. Forsyth
In Submission

2. **Denoising Upsampling Residual Autoencoders: Easy MAP Estimate for Image Enhancement**
A. Bansal, **A. Bhattad**, D. A. Forsyth
In Submission

1. **SIRfyN: Single Image Relighting from your Neighbors**
D. A. Forsyth, **A. Bhattad**, P. Asthana, Y. Zhong, Y. Wang
Technical Report

CONFERENCE PAPERS

5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
A. Bhattad, D. A. Forsyth
International Conference on 3D Vision (3DV), 2022

4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering (Best Paper Finalist)**
L. Wu , J. Y. Lee, **A. Bhattad**, Y. Wang, D. A. Forsyth
Computer Vision and Pattern Recognition (CVPR), 2022

3. **View Generalization for Single Image Textured 3D Models**
A. Bhattad, A. Dundar, G. Liu, A. Tao, B. Catanzaro
Computer Vision and Pattern Recognition (CVPR), 2021

2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
International Conference on Learning Representations (ICLR) 2020

1. **Improved Style Transfer with Calibrated Metrics**
M. Yeh*, S. Tang*, **A. Bhattad**, C. Zou, D. A. Forsyth (* for equal contribution)
Winter Conference on Applications of Computer Vision (WACV) 2020

Workshop Papers

2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
Computer Vision and Pattern Recognition (CVPR) Workshop 2019, ***Oral Presentation***

1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
A. Bhattad, J. Rock, D. A. Forsyth
Computer Vision and Pattern Recognition (CVPR) Workshop, 2018, ***Oral Presentation***

Students Mentoring

- Matthew Weston (PhD; present) | Project on Controllable Diffusion Models
- James Soole (MS; present) | Project on Illumination Physics in Dense Prediction Models
- Feiran Wang (MEngg; present) | Project on NeRF + Illumination Physics
- Liwen Wu (BS-MS; 2021-2022) | Paper at CVPR 2022 | Next position: PhD Student at UCSD
- Brian Chen (BS; 2020-2022) | Paper under review | REU at CMU's Robotics Institute
- Kexuan (Klaus) Zou (MS; 2019-2020) | Next position: Software Engineer at NVIDIA
- Anchu Zhu (BS; 2018-2019) | Next position: MS in CS at USC

Research Internships

Allen Institute of Artificial Intelligence, Seattle/Champaign(remote) *Feb'23 - present*
Research Intern with *Ani Kembhavi*
Project on Unified generative-discriminative model.

Intel ISL, Santa Clara/Champaign (remote) *Feb'21 - Aug'21*
Research Intern with *Stephan R. Richter*
Project on image-based object insertion.

NVIDIA ADLR, Santa Clara/Champaign (remote) *May'20 - Dec'20*
Research Intern with *Aysegul Dundar, Andrew Tao and Bryan Catanzaro*
Project on single image textured 3D inference via differentiable rendering (CP#3)

Fyusion Inc, San Francisco *May'19 - Aug'19*
Research Intern with *Abhishek Kar and Rodrigo Ortiz Cayon*
Project on image-based relighting of cars.

Siemens Corporate Technology, Princeton *May'17 - Aug'17*
Computer Vision & DL Intern with *Jan Ernst and Ziyan Wu*
Developed a semi-automated segmentation tool based on Polygon-RNN from CVPR 2017.

Computational Sensor Lab, Michigan State University *May'14 - July'14*
Research Intern with *Prof. Nizar Lajnef and Prof. Amir H. Alavi*
Built a framework for Damage Detection Using Self-Powered Wireless Sensors

Earthquake Engineering Research Center, IIIT Hyderabad *Jun'13 - July'13*
Research Intern with *Prof. Ramacharla Pradeep Kumar*
Designed Earthquake Resistant Multi-Storied Building

Teaching Experience

Computer Science, UIUC; *Graduate Teaching Assistant* *Jan'16 - Present*

- Applied Machine Learning (CS498 AML) in Fall'18 | Designed Kaggle competitions
- Data Structures (CS 225) in Spring'17 | Developed Assignments, Exams & Labs
- Introduction to Computing (CS 101) in Spring'16 & Fall'16 | **Rated as Excellent TA**

Other Projects

Pedestrian Detection and Automated Braking for Self-Driving Cars *Jan'17 - May'17*
Project Mentor: Prof. David A. Forsyth `[Video | Code]`

- Built a smooth autonomous braking system for pedestrian detection using YOLO & SSD

- Estimated pedestrian distance from perspective geometry using only a monocular camera
- Integrated the software built using Robotics Operating System and tested on a self-driving car

Services & Leadership

Organizer, CVPR-W on "Big Models Vs Scholars: How Should Academics Adapt?" June 2023

- Co-organizing a CVPR workshop to discuss the potential impact of large vision models on the Academic CV community
- Aims to bring fresh perspectives on redesigning collaborations, access to compute, and curricula.
- Positively shape a more inclusive, equitable, and diversity-rich future for computer vision via discussions and opinions through assorted panels and speaker talks

Organizer, External Speaker Series and Vision Lunch Jan 2021-Present

- Hosting the first external speaker series, inviting researchers outside of UIUC for technical talks
- Organizing computer vision reading and presentation group in the CS department of UIUC

Reviewer, Computer Vision Conferences Nov 2018-Present

- Serving as reviewer for CVPR, NeurIPS, ECCV and ICCV conferences and workshops.
- **Outstanding Emergency Reviewer** for CVPR 2021

Program Committee Jun 2019

- Adversarial Machine Learning in Real-World Computer Vision Systems, CVPR 2019 Workshop
- Security and Privacy in Machine Learning, ICML 2019 Workshop

Organizer, Vision Coffee, Urbana-Champaign *May'18 - Aug'18*

- Vision and Deep Learning paper/topic weekly discussion group
- Finalized agenda/topic, location and initiated discussions

Co-Founder, SchoolEngg *Apr'13 - May'15*

- Forum for high-school students; providing the firsthand knowledge of all engineering disciplines
- Currently, renamed as *PrepLift*, an education counselling start-up in India

Research Statement

Anand Bhattad, PhD Student, Computer Science, UIUC

My research focuses on two main themes: (1) incorporating illumination properties into existing models to enrich their capacity and (2) improving the robustness of current methods to allow for better generalization to unseen environments. These two themes are complementary, as both are necessary for creating systems that can handle a wide range of environmental conditions and extend the capabilities of current AI systems.



Figure 1: Our illumination-aware generative models offer precise relighting of scenes without altering their geometry or material properties. With applications in computer graphics, virtual and augmented reality, and robotics, it enables realistic and convincing lighting effects for diverse environments.

1 Illumination-aware Models

Incorporating illumination properties into existing models has been a primary focus of my research. Traditional methods that decompose images into albedo and shading often lead to unrealistic results, such as the loss of surface details or changes in the spatial layout of the scene. To address these challenges, I have developed a novel image decomposition technique that modifies the traditional albedo-shading decomposition by separating persistent maps, analogous to albedo and encoding materials and sharp edges, from transient maps, analogous to shading and encoding diffuse and specular shading without geometry information (surface normals).

This technique is based on the Retinex theory [9] and involves the use of Mondrian with fine edges to encode persistent maps and smooth Perlin noise containing white blobs to encode transient maps. The use of these specific patterns allows for the separation of image information into light-dependent and light-independent components, enabling the realistic manipulation of lighting in images and scenes without altering their albedo or geometry. By imposing decomposition consistency on image parts, this technique allows for the successful transfer of objects from one image to another without losing any surface details and enables the realistic relighting of complex scenes.

I have applied my novel image decomposition technique to develop two state-of-the-art (SOTA) object insertion methods: DIPR [5] and Illuminator [2]. Both methods are capable of abstracting scene lighting and decoding images with realistic novel object insertion into scenes, allowing for the successful transfer of objects from one image to another without losing any surface details. In addition to object insertion, I have also developed the first single image relighting method, StyLitGAN [3], which can generate realistic scenes with arbitrary lighting, including cast shadows, inter-reflections, and soft shadows without requiring any physical models or other CGI (computer-generated imagery) labeled data (Figure 1). This is a significant improvement over previous meth-

ods, which often require extensive physical modeling or rely on large amounts of labeled data to produce similar results [6]. StyLitGAN can achieve these results by preserving scene semantics, ensuring that the resulting images are not only photorealistic but also semantically accurate.

My research on enhancing AI capabilities with illumination properties has significant implications for a wide range of industries and applications. This technique can be applied to create new types of content, such as virtual advertisements that can be inserted into live video streams or dynamic lighting for video games. In the realm of augmented and virtual reality experiences, my research offers exciting possibilities for enhancing the user experience. By enabling more realistic and convincing virtual environments, my techniques can help to create more engaging and immersive experiences for users. This has implications for a wide range of applications, including CGI for film, video games, training simulations, education, and entertainment.

2 Improving Model Performance through Robustness and Generalization

One of the major challenges in several SOTA models is the lack of robustness to changes in lighting conditions and equivariant properties. To address this, I developed an equivariant regularization technique [4] that ensures the model's predictions are consistent even when the input image is transformed, resulting in significantly improved accuracy, and robustness to lighting changes.

In addition to improving robustness, I also focused on enhancing the view generalization capabilities of generative models for novel view synthesis. One approach that I built involves using a controllable deformation model to improve the inferred underlying 3D geometry and texture of an object from a single 2D image [8]. Another method, called DIVeR, uses deterministic estimates of the volume rendering integral to learn accurate and realistic 3D object models from small numbers of images, improving the robustness of Neural Radiance Fields [7]. These methods significantly improve novel view synthesis, making them more capable of handling challenging conditions such as occlusions, and thin and translucent surfaces; reducing the risk of failure in critical applications.

Another limitation of current models is the trade-off between image resolution and model complexity, resulting in increased computation time and the risk of overfitting. To address this issue, I have developed Denoising Upsampling Residual Autoencoder (DURA) [1], a denoising autoencoder that generates high-resolution images in less than a second without noticeable artifacts, and can be applied to a wide range of image-denoising tasks and imaging applications.

My work on improving the robustness and generalization of AI systems through equivariant regularization and denoising autoencoder models, as well as my techniques for improving the view generalization capabilities of generative models, has shown great promise in addressing key challenges in the field, offering a comprehensive suite of techniques to improve the reliability, efficiency, and effectiveness of AI systems in a variety of applications.

3 Future Research

Moving forward, my research will continue to focus on developing advanced models that can generate realistic and detailed graphics with faster rendering times and handle complex scenes effortlessly. By integrating both generative and robust discriminative properties in a unified model, I aim to enable comprehensive scene understanding and content creation for a wide range of applications. Ultimately, my work will contribute to the development of more reliable and efficient AI systems that can handle challenging conditions and reduce the risk of failure in critical applications, making a significant impact on the safety and reliability of real-world technologies.

References

[1] A. Bansal, **A. Bhattad**, and D. A. Forsyth, "Dura: Denoising upsampling residual autoencoders for image enhancement," *Under Review*, 2023.

[2] **A. Bhattad**, B. Chen, S. Wang, S. R. Richter, and D. A. Forsyth, "Illuminator: Learning to correct lighting without ground truth data," *Under Review*, 2023.

[3] **A. Bhattad** and D. A. Forsyth, "Stylitgan: Enriching stylegan with illumination physics," *Under Review*, 2023.

[4] Y. Zhong, **A. Bhattad**, Y.-X. Wang, and D. A. Forsyth, "Learning equivariant dense prediction networks," *Under Review*, 2023.

[5] **A. Bhattad** and D. A. Forsyth, "Cut-and-paste object insertion by enabling deep image prior for reshading," in *Proceedings of the International Conference on 3D Vision*, 2022.

[6] Z. Li *et al.*, "Physically-based editing of indoor scene lighting from a single image," *arXiv preprint arXiv:2205.09343*, 2022.

[7] L. Wu, J. Y. Lee, **A. Bhattad**, Y.-X. Wang, and D. Forsyth, "Diver: Real-time and accurate neural radiance fields with deterministic integration for volume rendering," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16 200–16 209.

[8] **A. Bhattad**, A. Dundar, G. Liu, A. Tao, and B. Catanzaro, "View generalization for single image textured 3d models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 6081–6090.

[9] E. H. Land and J. J. McCann, "Lightness and retinex theory," *Josa*, vol. 61, no. 1, pp. 1–11, 1971.

Title of the Poster:
StyLItGAN: Prompting StyleGAN to Generate Images with New Illumination Conditions

Authors: Anand Bhattad and David A. Forsyth

UNIVERSITY OF ILLINOIS
AT URBANA-CHAMPAIGN

D.A. Forsyth
Professor

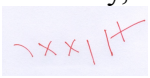
Department of Computer Science

201 N. Goodwin
Urbana, IL 61801

To whom it may concern,

This confirming that Anand Bhattad, who is applying to attend the CVPR 2023 Doctoral Consortium, will likely graduate Summer 2023. For the record, I strongly support his application.

Sincerely,



D.A. Forsyth,
Professor

Background: I am currently Fulton-Watson-Copp chair in computer science at U. Illinois at Urbana-Champaign, where I moved from U.C Berkeley, where I was also full professor. I have occupied the Fulton-Watson-Copp chair in Computer Science at the University of Illinois since 2014. I have published over 170 papers on computer vision, computer graphics and machine learning. I have served as program co-chair for IEEE Computer Vision and Pattern Recognition in 2000, 2011, 2018 and 2021, general co-chair for CVPR 2006 and 2015 and ICCV 2019, program co-chair for the European Conference on Computer Vision 2008, and am a regular member of the program committee of all major international conferences on computer vision. I have served six years on the SIGGRAPH program committee, and am a regular reviewer for that conference. I have received best paper awards at the International Conference on Computer Vision and at the European Conference on Computer Vision. I received an IEEE technical achievement award for 2005 for my research. I became an IEEE Fellow in 2009, and an ACM Fellow in 2014. My textbook, "Computer Vision: A Modern Approach" (joint with J. Ponce and published by Prentice Hall) is now widely adopted as a course text (adoptions include MIT, U. Wisconsin-Madison, UIUC, Georgia Tech and U.C. Berkeley). My recent textbook, “Probability and Statistics for Computer Science”, is in the top quartile of Springer computer science chapter downloads. A further textbook “Applied Machine Learning” has just appeared in print. I have served two terms as Editor in Chief, IEEE TPAMI. I serve on a number of scientific advisory boards.

telephone 217-333-3426 *fax* 217-333-3501 *email* admin@cs.uiuc.edu

Potential Mentors

1. Richard Szeliski szeliski@cs.washington.edu
2. Shree K. Nayar nayar@cs.columbia.edu
3. Michael Black black@tuebingen.mpg.de
4. Erik G. Learned-Miller elm@cs.umass.edu
5. Ira Kemelmacher-Shlizerman kemelmi@cs.washington.edu
6. Phillip Isola phillipi@mit.edu
7. Philipp Krähenbühl philkr@cs.utexas.edu